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SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct
I ASWATHA . V - 192525390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

CO2 AT 3

1) Auto-scaling & load balancing in cloud computing

Introduction:

Cloud computing helps companies manage changing workloads by providing flexible computing resources. Auto-scaling & load balancing are important techniques used to handle sudden increases in website traffic

Auto-Scaling:

→ Automatically increases or decreases cloud resource based on demand.

→ During seasonal scales, servers are added to handle

→ Traffic ↓, unnecessary resources are removed

→ Maintain performance while controlling operational costs.

Load Balancing:

→ Distributes incoming user requests across multiple server.

→ prevents a single server from becoming overloaded.

→ If one server fails, traffic can be redirected to healthy servers

→ Improves application availability & response time.



Role of Monitoring Tools:

→ Tracks CPU usage, memory, traffic etc.
→ Identify performance issue & server failures.
→ Alerts help administrator respond quickly to problems.
→ Monitoring data helps auto-scaling systems decide when resources are needed

Benefits:

→ Reduces downtime during traffic
→ Improves website speed & user experience.
→ Supports a large no. of users simultaneously.
→ Reduce cost by avoiding unecessary resources.

Conclusion:

Auto-scaling & load balancing reduce downtime, improve user experience, & control cloud costs.

2) Cloud Storage for Secure Backup & Disaster Recovery:

Introduction:

Cloud storage allows organizations to store large amounts of data securely. It supports backup, recovery & cost-effective data management.

Data Durability & Redundancy:

→ Durability protects data against loss or damage.

→ Reduadancy maintains multiple copies of data

→ Copies may be stored accross different devices or locations.

→ This helps recover data after hardware failures.

Encryption & Access control:

→ protects data during storage & transfer.

→ Access control allows only authorized users to access files

→ Multi-factor authentication provides additional security.

Advantages:

→ Reduces the need for purchase physical devices.

→ Pay-as-you-go

→ Hardware maintenance costs is less

→ lower-cost option for old backups.

Conclusion:

Cloud storage provides secure, reliable, affordable backup facilities while supporting faster disaster recovery.

3) Content Delivery Networks & Global Cloud Deployment.

Introduction:

Companies use CDNs and global cloud infrastructure to deliver applications quickly to user across different countries.

CDN - Content Delivery Network

→ It consists of servers distributed across different locations.
→ Stores cached copies of image, videos etc.
→ users receive content from servers
→ This reduces latency & loading time.

Role of Edge Sowers:

→ Deliver content closer to user
→ Reduce Network delays
→ Improve video playback & response
→ Reduce the workload on central server.

Global Availability:

→ cloud providers operate data centers in multiple regions
→ Applications can serve user from diff location
→ Replication & backup availability.



Conclusion

CDNS, edge servers, global cloud regions improve application speed, reliability & user experience.

4) Cloud security & Regulatory compliance.

Introduction :

Financial institutions handle sensitive information such as customer details & transaction records. Cloud security features help protect this data and support requirements.

Regulatory compliance & Auditing :

→ Cloud providers offer security controls & compliance certifications.

→ Audit reports help verify security practices.

→ Regular audits identify security gaps.

Logging & Monitoring :

→ Logging records login, transactions & data access

→ Monitoring detects suspicious activities.

→ Alerts help administrators respond to threats

→ Audit logs support security investigation.







Role of Encryption :

→ protects data during storage & transfer.

→ prevents unauthorized users from easily reading sensitive information.

→ Secure key management limits access to data

Data Sovereignty :

→ Data must be stored & processed according to applicable laws.

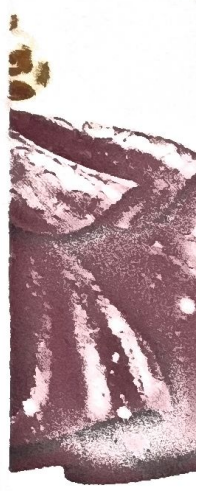
→ Organizations should select suitable cloud regions

→ Backup locations & data access must be checked.

Conclusion :

Encryption, monitoring, auditing & proper data location managment help financial institutions maintain secure & compliant cloud services.





5) Containerization & Orchestration in Cloud.

Introduction :

Containerization package applications & their dependencies into lightweight units called containers. It simplifies application deployment & managment.

Containers VS VM

Features	Containers	Virtual Machines
OS	Share host OS kernel	Separate guest OS
Resource	Low	high
Startup	faster	Slower
Portability	Compatible platforms	compatible hypervisor.

Orchestration :

→ Kubernetes manages containers across servers

→ Restarts failed containers when configured to do so.

→ Scales application instances based on demand.

→ Simplifies deployment & updates.

Benefits :

→ Support consistent deployment

→ Allows applications to run on cloud platform

→ Reduces deployment errors.

Conclusion :

Containers provide lightweight & portable, while Kubernetes simplifies scaling & management across cloud environment.